

Chapter1

Artificial Intelligence

The artificial intelligence (AI) world is quickly growing as it reaches several different industries. You currently will see AI used for many different purposes from agriculture to automotive; with the passage of time you will see even more of it. Agriculture is one of the most important branches of the IT industry. Agriculture is an important industry and is a big part of our economy's base. In terms of annual sales, the agricultural industry contributes to our economy almost 17%. AI is becoming a technical advancement that boost and protects crop yield, as climate change and populations rise. Here in this chapter basics of Artificial Intelligence have been discussed how it can be related to agriculture.

1. Introduction

Artificial Intelligence has become most prominent in current generation. Every field is utilizing Artificial Intelligence for their better results and which also result in saving a lot of time. As Artificial intelligence is boosting different sectors to increase the productivity and efficiency and its utilization across different sectors like in the field of CIVIL Artificial Intelligence can be used to analyze the structures of buildings, dams etc., and Artificial Intelligence solutions are assisting in so many ways to overcome the traditional challenges in every field.

I. What is Artificial Intelligence

Artificial Intelligence is one of the most prominent field of Computer Science Engineering which attempts to redefine the tasks which are carried out by humans in their daily life with better results and maximum accuracy. Artificial Intelligence is actually meant to ease the work of human where the word artificial states the work done without any human interference and intelligence states the capabilities like human brain to perform tasks. The field artificial Intelligence has given so many definitions from time to time which can be typically stated as

1. The Engineering and Science come together to make machine Intelligent.
2. To make a machine intelligent which can imitate the human behavior
3. The branch of Computer Science which can perform human like activities which can include thinking, learning etc.,
4. An area of Computer Science which makes machine to act and behave like humans.

Artificial intelligence enables and increase the machine capabilities and makes them to comprehend facts, events, analyze the situations and understand to derive the knowledge from surrounding and to act upon them to solve the problem which can

lead to better results than humans. The primary focus of Artificial Intelligence is always been to develop a machine which can psychological and to simulate the behavior of humans through constructing the thoughts like human from any surrounding or situations.

II. Introduction of Artificial Intelligence in Agriculture

Agriculture has been one of the oldest professions in the world. Humanity from their early ages to current generation has come to a long way in development of agricultural practices and its techniques in production of food humanity consumes in their daily lives. As it has seen from past generations to till current generation there always been an increase in population which leads to increase in consumption of food and decrease in production of it by 20%.

As Agricultural Production is decreasing day by day and from generation to generations, Artificial Intelligence can help us to automate the production of food crops in agricultural fields without any human interference which will lead into increase of food production. Artificial Intelligence solutions can perform tasks like analyzing the field for crop harvesting, monitoring of crops and soil conditions from time to time, it can forecast the weather and predict it for future purpose etc., using artificial intelligence in agriculture can leverage the assistance to the farmers and improve their farming techniques by serving them with correct data as per requirement. Later on discuss about types of AI and its different branches which can have a huge impact on Agriculture.

III. Stages and types of Artificial Intelligence

Artificial intelligence has different stages which can also be said to types of artificial intelligence which are

1. Artificial Narrow Intelligence
2. Artificial General Intelligence
3. Artificial Super Intelligence

These three types can said to be how AI can evolves, and now lets see each of them separately

Artificial Narrow Intelligence (ANI): It can also be called as weak AI, ANI can perform only the narrow-defined tasks which don't have cognitive and thinking ability, these are just the predefined functions which are already feeded into the system.

For Example, Siri, Alex, Self-Driving cars, Alpha Go, Sophia the humanoid etc., are the ANI we have developed so far.

Artificial General Intelligence (AGI): it can also be called as strong AI, AGI is the second stage of evolution of Artificial Intelligence which posses the ability to think and make decisions just like human. Currently we haven't developed such system which can said to the perfect example for artificial General Intelligence system.

Strong AI is by far considered as threat to human existence by so many researcher's and scientist and even by technology giants like Google, Microsoft, Tesla etc., The

complete development of an Artificial Intelligence System could spell the end to an humanity and by far human is not capable of competing such system which is most powerful in terms of thinking and intelligence.

Artificial Super Intelligence (ASI): It can be stage of Artificial Intelligence where the capacity of AI system will pass all the limitations and can suppress the human beings. ASI is current now only a hypothetical situation which can happen in near future if mankind continues their advancement in development of Artificial Intelligence.

As we have seen the stages of evolution of artificial intelligence let see some of the artificial intelligence system which are purely based on the functions they perform. They are

1. **Reactive Machine Artificial Intelligence:** This type of AI system takes the input as current data and perform the task based on current situations. Reactive AI machine system cannot analyze and predict the future task or their actions of any particular situation. These systems always perform from narrow range predefined tasks.
2. **Limited Memory AI:** These AI system mostly based on the data they have stored and usually perform the task by studying the previously performed task and also perform the prediction of future task by the machine.
3. **Theory of Mind AI:** These systems are more advanced type of AI which is used to speculate and play a major role in terms of psychology. These AI systems always performs the tasks related to Emotional Intelligence and the thoughts which can be comprehended in any situation of an individuals.
4. **Self-Aware AI:** These System so far most advanced and fall under last stage of artificial Intelligence where AI system consist of consciousness and become self-aware of everything. In the nearby future these can be achievable.

As we seen the different stages and types of Artificial Intelligence system, but apart from this these systems are made by using several technologies which are considered to be as a branch of artificial intelligence. They are

- Machine Learning
- Deep Learning
- Computer Vision
- Data Science
- Big Data

These branches of artificial Intelligence have been massively utilized in agriculture filed to improve production and bring efficiency in harvesting. We will discuss these things in later chapters briefly and their implementations with some projects.

IV. Applications of AI in Agriculture

Artificial Intelligence has many applications and can be used in so many ways in the field of agriculture. As Agricultural fields provides us so much data in terms of temperature, precipitation, solar radiation and wind speed which can be analyzed and

help us to benefit in terms of harvesting from agricultural fields. The best part in implementing artificial intelligence in agriculture is that it will not remove the jobs of farmers but it will help the farmers to improve their process of harvesting in the fields.

Artificial intelligence can have best application in agriculture, some of them are like:

1. **Artificial Intelligence Agriculture Bots:** AI Enabled Agriculture bots can assist the farmers to find the more efficient ways to find better harvesting techniques in any type of agricultural field and guide the farmer to protect their crops from weeds. These bots can also help in the agriculture fields in harvesting the crops at larger volume and at faster pace than any human laborer's in this way farmers can harvest more crops in less time. By implementing computer vision with the bots, it can also detect the several diseases into the crops and can spray pesticides at correct time.
2. **Monitoring crop and soil health:** Artificial Intelligence can be utilized in an efficient way to monitor and identify the defects and nutrients deficiencies in the soil. With the help of computer vision's image recognition approach systems can be developed which can identify the defectness in crop and using the deep learning applications flora patterns can be analyzed in agriculture. These type of AI Enabled applications will make easy to understand the different parts of crops, their defects, plant pests and its disease.
3. **Agricultural Expert System:** Expert System can be best for farmers in providing proper guidance. Expert System are by default are the system which have expert level knowledge stored in them which is being gathered from different experts of that field and which is based on different situations and practices which are being performed by experts.
4. **Forecasted Weather data:** Artificial Intelligence in an advance way can help the farmers to make them updated regarding the sudden change in weather which will help them to protect their crop from hazardous situation. The Analysis done by the system can help the farmers to take precautions (need to think from here).
5. **AI with Drones:** Drone Technology helps the farmers to improve their crop yields and reduce the cost. User pre programs the drone's route and once it is deployed into the field which can have the aerial technology enabled with various features like capturing live image data of crops, spraying pesticides on disease detection at particular crops in the field etc., Drones can be programmed for any particular tasks too related to the agricultural field like analyzing the field from time to time and scanning it for any infected or diseased crops. Camera's in the drone which are enabled with imaging technology can help with overall field management by providing real time crop requirement of water, fertilizers, pesticides etc., Machine learning can be used in the proper analysis of crop and soil condition by providing us proper insight of crop and its requirements which can increase the production of crop on the field.

6. **Indoor Harvesting with AI:** Indoor harvesting has become a trend in current time where people can harvest different crops in small area by providing proper artificial lighting according to its requirements. By Using Artificial Intelligence, indoor harvesting will become more simpler and easier because it can analyze everything in a closed environment and can automate the process of harvesting either it can be hydroponics or aquaponics.

V. Advantages of AI in Agriculture

Artificial Intelligence has many advantages in different field and here it has been discussed about the advantages in agriculture field which are

- Artificial Intelligence (AI) has provided us more efficient methods to produce, harvest and can guide us to sell essential crops by analyzing the market values.
- AI can be implemented to check crop defectiveness and to improve harvest of healthy crops.
- The increase in technology utility based on artificial Intelligence can strengthen the agriculturally based business and markets to run more efficiently.
- AI can help to improve the crop management practices and its harvesting.
- AI based solutions can be used to solve the various traditional challenges which are being faced by farmers like climate change, an infestation of pests and weeds that reduces yielding of food crops.

Summary

As so far discussed in this chapter regarding artificial Intelligence and its types narrowing down its advantages in agriculture. It is also discussed that artificial Intelligence can lead to increase in yield and protection of crops while practicing agriculture in the fields in real time which can be very useful to farmers nowadays. Due to advancement in technology farmers should also an advancement in agricultural techniques to increase their crop production and saving time in which artificial Intelligence can play a very big role.